Finding Evidence Is Solved, Combining It Is Not: A Fusion-Depth Analysis of Question Answering over Digital Product Passports

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***Abstract*—A Digital Product Passport is a machine-readable record of a product: what it is made of, how it performs, and how it must be handled at end of life. Regulators and auditors need to ask questions of these records, and the answers must be defensible. We study such question answering as a problem of *data fusion*, meaning the task of combining information from more than one source into a single answer. We separate questions by how much fusion they need. A *recall* question is answered by one stated value. An *open* question must be assembled from several parts of the record. A *schema-governed* question cannot be read off at all: it needs a value from the record together with the product schema that gives that value its meaning, such as deciding whether a stated component belongs to a declared class. On a verified set of 6,270 questions across 3 industrial domains, evaluated on identical rows, three published configurations show a trade-off rather than a ranking. A long-context model and Logic-LM answer 99.93% and 99.96% of recall questions but fall to 93.09% and 93.62% on rule-governed ones; LINC, which routes every question through a formal representation, is the best of the three on rule-governed questions at 95.36% yet the worst on the other two depths. No configuration is strong at both ends. The headroom left to a perfect score rises monotonically with depth, from 0.04 to 2.18 to 4.64 points. A configuration that applies symbolic validation selectively, and withholds an answer when the combined evidence is insufficient, answers all 6,270 correctly, and across all three baselines there is no question a baseline answers correctly and it does not. On 4,021 queries over an unseen public product database it reaches 97.21% decision accuracy at 73.0% coverage with a calibration error of 0.021, lowering coverage rather than guessing where the record is sparse. The reading we draw is that fusion should be matched to the depth a question needs: uniform strategies pay for it at one end or the other.**

***Index Terms*—intelligent data fusion, digital product passport, question answering, evidence combination, selective prediction, benchmark construction**

I. Introduction

A Digital Product Passport, or DPP, is a structured record that describes a product across its life: its materials, its compliance evidence, how it can be repaired, its environmental indicators, and what should happen to it at end of life. European product policy is making such records a standard way to disclose this information [1]. The people who read them are often auditors, regulators, or procurement staff, and their decisions have to hold up when challenged.

Asking questions of such a record is not one task but several, and they differ in how many pieces of information must be brought together. Some answers sit in the record as a single stated value. Some must be assembled from several places. Some cannot be read off at all: they follow from a value in the record combined with the schema that defines what that value is — the class hierarchy a component belongs to, or the vocabulary a field is drawn from — which lives outside the record itself. We call the number of kinds of source an answer must draw on its *fusion depth*.

The distinction matters because existing systems commit to one fusion strategy and apply it to every question. Retrieval-augmented generation, in which a system fetches relevant passages and then answers from them, combines evidence loosely and leaves the reasoning to the language model. Long-context models, which read the whole record at once, do the same with a larger window. Neuro-symbolic systems go the other way and translate every question into a formal statement for a solver. Each strategy suits one depth.

Our central observation is that this shows up as a trade-off, not a ranking. On identical questions, the configuration that is best at the deepest level is the worst at the shallowest, and vice versa. A single accuracy figure hides this completely, because the three published systems we compare are within 2.11 points of each other overall. Split by depth, they are not close at all, and the amount of accuracy still available rises monotonically with depth.

We report a controlled comparison, not a system description. The architecture of the configuration that does best here is reported separately; this paper treats it as one point in a design space and describes it only as far as the comparison requires.

Contributions.

- A fusion-depth analysis of product-record question answering, showing a trade-off among published systems that a single accuracy figure conceals, and headroom that grows monotonically with depth.

- A verified evaluation set of 6,270 questions in which every excluded row is removed for an objective reason fixed before evaluation, so no configuration is scored on a question whose correct answer is genuinely ambiguous.
- A paired comparison on identical rows showing a one-sided result: selective symbolic validation costs nothing anywhere on this set, which is what distinguishes it from the uniform strategies.
- A generalisation arm on 4,021 queries over a public database with a different schema, where the system is measured on whether it knows when the record cannot support an answer.

II. Related Work

Retrieval-augmented generation answers a question by first fetching passages and then generating from them. Its evaluation normally reports how often the answer is right when an answer exists, and says little about whether the system can tell that no answer exists.

Long-context models take the opposite approach and read the whole record. This helps, but evidence placed away from the start or end of a long input is often underused [2], so a longer window is not by itself a fusion mechanism.

Neuro-symbolic question answering translates a question into a formal statement and hands it to a solver. LINC [3] and Logic-LM [4] are the two systems we compare against. Both were built for logical-reasoning benchmarks rather than for product records, and part of what this comparison measures is how well that transfers.

Selective prediction is the study of when a system should answer and when it should stay silent [5]. It supplies the coverage and risk measures we use. For the decision to mean anything, the system's confidence must be calibrated, that is, a stated confidence of 90% should be right about 90% of the time [6].

What we add is the grouping. Prior comparisons report one accuracy figure per system, which presumes the systems can be ranked. Reporting accuracy against fusion depth shows that on this benchmark they cannot: each is best somewhere and worst somewhere else, and the single figure averages that away.

III. Fusion Depth and Evaluation Design

A. Three depths

We group questions by how many sources an answer must combine.

Depth one, recall (2,750 questions). The answer is a value stated in the record. The task is to locate it. Nothing has to be combined.

Depth two, open (871 questions). The answer is not stated in one place and must be put together from several parts of the record. Sources are combined, but they are all of the same kind.

Depth three, schema-governed (2,649 questions). The answer is not asserted anywhere in the record. It follows from a value the record does state, combined with the schema that classifies it — for example, whether a named component is an instance of a declared product class. Two different kinds of source must be combined, and reading the record correctly is necessary but not sufficient.

We state the scope of this class precisely, because it is narrower than the regulatory-compliance questions a product-passport setting might suggest. Every question in it asks about class membership under the schema; none applies a numerical threshold or a compliance limit. A further 173 of the 2,649 concern the schema vocabulary itself rather than product content, and Section V reports what changes if they are excluded.

The grouping is a property of the question, not of any system, and it is fixed before evaluation. It is deliberately coarse: it counts kinds of source, not reasoning steps, so that it can be assigned without judging difficulty. The depths are also not evenly represented, which is itself a property of the setting — regulated product records generate a large share of rule-governed questions, and that is precisely where the published systems diverge.

B. Why a verified evaluation set

A benchmark built from real product documentation contains questions whose correct answer is not uniquely determined. A knowledge base may support two conflicting statements about the same entity, a document label may be malformed, or the expected value may appear only as a placeholder. Scoring any system on such a question measures the benchmark rather than the system.

We therefore keep a question only when its answer is uniquely and atomically supported by the evidence. The exclusion criteria are objective, applied without looking at any system's output, and fixed in advance. 645 rows are removed under 8 recorded reasons, the largest groups being conflicting knowledge-base evidence and malformed document labels. What remains is 6,270 questions across 3 industrial domains.

The exclusions are published with the benchmark, so a reader who disagrees with a criterion can recompute without it.

C. Configurations

All four configurations are scored on the same 6,270 rows. The external baselines are scored from their released outputs filtered to those rows, not re-run.

LongCtx places the record in a long-context language model. *LINC* and *LogicLM* translate the question into a formal representation and delegate to a solver. The fourth configuration adds two things: it checks rule-governed questions against a machine-readable version of the applicable constraints, and it withholds an answer when the combined evidence is not sufficient to support one.

IV. Results

A. A trade-off, not a ranking

Table I and Fig. 1 report accuracy by depth.

The overall column is nearly uninformative: the three published systems sit within 2.11 points of each other, which

TABLE I

ACCURACY ON THE SAME 6,270 VERIFIED QUESTIONS, GROUPED BY FUSION DEPTH. BEST BASELINE PER COLUMN IN BOLD. THE THREE PUBLISHED SYSTEMS ARE WITHIN 2.11 POINTS OVERALL, BUT NO ONE OF THEM IS BEST AT MORE THAN ONE DEPTH.

	by fusion depth			
configuration	recall	open	rule	overall
LongCtx	99.93	96.56	93.09	96.57
LINC	95.45	91.62	**95.36**	94.88
LogicLM	**99.96**	**97.82**	93.62	96.99
headroom to ceiling	0.04	2.18	4.64	
fused + selective	100.00	100.00	100.00	100.00

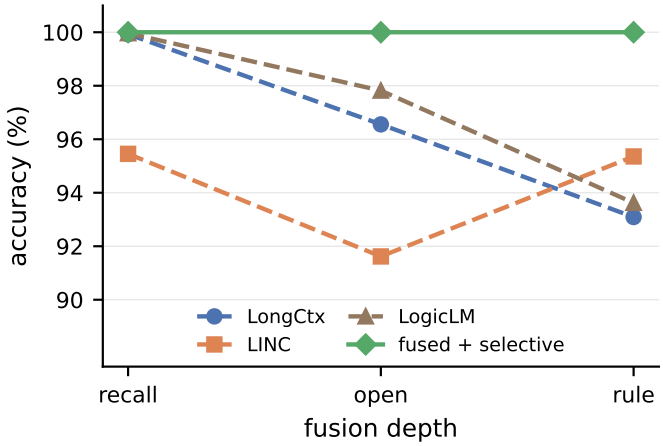


Fig. 1. Accuracy against fusion depth. The lines cross. Systems that combine evidence loosely are strong where the answer is a stated value and weak where a rule must be applied; the system that formalises every question is the reverse.

would ordinarily be read as three comparable systems. Split by depth, they are not comparable at all.

LongCtx and LogicLM find stated values almost perfectly, at 99.93% and 99.96%, and then fall to 93.09% and 93.62% once a rule must be applied. The evidence they need is in front of them; what fails is combining it with a constraint that is not in the record.

LINC behaves in the opposite way. It is the strongest of the three on rule-governed questions at 95.36%, which is what its design targets, but the weakest on recall at 95.45% and on open questions at 91.62%. Because it translates every question into a formal representation before answering, it applies deep-fusion machinery to questions that need none, and pays for it. Depth-three capability bought this way is not free.

That is the finding we consider most useful: on this benchmark, being good at one depth has come at the cost of another. The last row of Table I gives the headroom, the accuracy still available to the best baseline at each depth. It rises monotonically, from 0.04 points at depth one to 2.18 and then 4.64. The remaining errors are not retrieval errors, and they are not distributed evenly.

Table II gives one worked example per class, quoted from the released outputs. What the aggregate numbers cannot show is that none of these failures looks like a failure. Asked what warranty standard a heating record declares, LINC answers *EU F-Gas Regulation*: a real regulation, correctly spelled, and not what the record says. Asked who the EU authorised representative is, it answers with the manufacturer's legal name rather than the declared value *Same as manufacturer* — it resolved the reference instead of reporting the field. At the reasoning class the direction reverses, and the two systems that combine evidence loosely answer from plausibility: asked whether *Gas* is an entity in the product schema, both decline it, while the schema itself asserts that it is.

The last row extends the pattern to a question with three sources at once: a value the user set earlier in the session, a compliance constraint, and a field from a passport document. Both baselines recover two of the three and then agree on the same wrong answer, replacing the required standard with *UN 38.3* — again a real standard, and the wrong one here. This row comes from a separate compositional evaluation rather than the verified subset, and LINC was not run on it; we report it as an illustration, and it contributes no number to any table.

The common shape is that each wrong answer substitutes the system's own prior for what the record states, and does so fluently. This is why answer-level review does not catch these cases in practice, and why the paired design in Section IV-B matters more than the averages.

Table III shows this is not an artefact of one domain. Ordering the three domains by the share of rule-governed questions they carry, from 53.9% down to 33.3%, baseline accuracy falls in the same order. With 3 domains this is an ordering rather than a correlation, and we report it as such.

B. Selective fusion pays no penalty here

The trade-off above is what a selective strategy is meant to avoid: apply symbolic combination to the questions that need it, and leave the others alone. Whether that works cannot be settled by average accuracy, because a system can improve on average while losing cases elsewhere, which is exactly what the uniform strategies do. Since every configuration is scored on identical rows, the comparison can be paired instead: for each question we can ask which configurations got it right. Table IV counts the disagreements in both directions.

The baselines leave 215, 321, and 189 errors on these rows; the selective configuration leaves 0. Each comparison is significant by an exact McNemar test at $p < 10^{-56}$, but the zeros in the right-hand column say more than the p-values do. Where LINC bought rule-governed accuracy by giving up recall accuracy, applying symbolic validation selectively gave up nothing measurable on this set. That is the difference between a depth-aware strategy and a uniform one, and it is visible only because the comparison is paired.

C. Knowing when the sources are not enough

Deciding whether to answer is itself a fusion question: it asks whether the combined evidence is sufficient. To test this away from the documentation the configuration was developed on, we evaluate 4,021 queries over Open Food Facts [7], a

TABLE II

ONE WORKED EXAMPLE PER QUESTION CLASS, QUOTED FROM THE RELEASED PER-QUESTION OUTPUTS. A TICK MARKS THE RECORDED VERDICT; WRONG ANSWERS ARE SET IN ITALIC. NONE OF THESE FAILURES IS A FLUENCY FAILURE: EVERY WRONG ANSWER IS WELL FORMED AND PLAUSIBLE, AND EACH SUBSTITUTES THE SYSTEM'S OWN PRIOR FOR WHAT THE RECORD STATES. ANSWERS IN THE DEPTH-AWARE COLUMN ALSO CARRY THE IDENTIFIER OF THE RECORD THEY WERE DRAWN FROM, OMITTED HERE FOR WIDTH. THE FIRST THREE ROWS ARE FROM THE VERIFIED SUBSET; THE FOURTH IS FROM A SEPARATE COMPOSITIONAL EVALUATION, ON WHICH LINC WAS NOT RUN. THAT ANSWER HAS THREE COMPONENTS AND THE SYSTEMS DIFFER ON EXACTLY ONE, SHOWN IN BOLD, SO THE CELL LISTS THE THREE COMPONENTS RATHER THAN THE FULL SENTENCE.

question (gold in italics)	LongCtx	LINC	LogicLM	depth-aware	verdict
basic According to viessmann_seed_0017, what is the Standard? *2 years parts & labour*	✓ 2 years parts & labour	× *EU F-Gas Regulation*	✓ Standard: 2 years parts & labour	✓ 2 years parts & labour	LINC swaps the declared field for a real regulation
open According to viessmann_seed_0002, what is the EU authorised representative? *Same as manufacturer*	✓ Same as manufacturer (sample)	× *Viessmann Climate Solutions SE*	✓ Same as manufacturer (sample)	✓ Same as manufacturer (sample)	LINC answers from world knowledge, not from the record
reasoning Is Gas a Thing? *Thing*	× *no*	✓ yes	× *"Gas"*	✓ Yes. Gas is a Thing.	both answer from plausibility rather than the schema
multi-source Using my saved preference and both product records: give my preferred supplier for ProductB, one compliance… *PrintPack — RoHS — separate collection symbol…*	× *PrintPack · **UN 38.3** · separate collection symbol*	*not run*	× *PrintPack · **UN 38.3** · separate collection symbol*	✓ PrintPack · **RoHS** · separate collection symbol	each recovers two of three sources, then substitutes a real but wrong standard

TABLE III

ACCURACY BY INDUSTRIAL DOMAIN, ORDERED BY THE SHARE OF RULE-GOVERNED QUESTIONS EACH DOMAIN CARRIES. BASELINE ACCURACY FALLS AS THAT SHARE RISES; THE ORDERING IS RECOMPUTED FROM THE ARTIFACTS ON EVERY BUILD.

configuration	battery	lexmark	viessmann
rule-governed share (%)	53.9	45.7	33.3
LongCtx	93.70	96.82	97.75
LINC	93.77	92.56	97.60
LogicLM	94.25	97.56	97.79
fused + selective	100.00	100.00	100.00

TABLE IV

PAIRED DISAGREEMENT. THE RIGHT-HAND COLUMN IS ZERO FOR EVERY BASELINE: NO BASELINE ANSWERS CORRECTLY A QUESTION THE FUSED CONFIGURATION MISSES.

baseline	fused-only correct	baseline-only correct
LongCtx	215	0
LINC	321	0
LogicLM	189	0

public product database with different field names, a different schema, and a great deal of genuinely missing data.

Accuracy here is *decision* accuracy: answering correctly when the record supports an answer, and staying silent when it does not. The configuration reaches 97.21% (Wilson 95% interval 96.66–97.68%) while answering 73.0% of queries. Its calibration error is 0.021 and the area under its risk–coverage curve is 0.025. When it does stay silent it is almost always right to: abstention precision is 0.998 at recall 1.000.

TABLE V

PUBLIC-SCHEMA ARM BY QUERY TYPE. WHERE THE UNDERLYING FIELDS ARE SPARSELY POPULATED THERE IS LITTLE TO COMBINE, AND THE SYSTEM LOWERS COVERAGE RATHER THAN GUESSING.

query type	n	accuracy	coverage	ECE
nutrition	1450	100.00	69.6	0.004
missingness	367	92.92	100.0	0.071
attribute	1146	99.83	99.8	0.002
sustainability	1039	91.92	38.0	0.060
fairness	19	100.00	100.0	0.000
all	4021	97.21	73.0	0.021

Table V shows the behaviour is not uniform across the 5 query types. Questions about stated nutrition values and product attributes are answered essentially perfectly, which is depth one on a new schema. The hardest type is sustainability, at 91.92%, and there the system answers only 38.0% of the time. Those fields are sparsely populated and inconsistently defined across the database, so there is often little to combine, and the system responds by declining rather than guessing. Its calibration error rises on the same slice, to 0.060, which is the behaviour we want: it is least confident exactly where it answers least often.

V. WHAT THIS SHOWS, AND WHAT IT DOES NOT

A perfect score invites suspicion, and it should. Four qualifications matter.

The trade-off is observed across three published systems on one benchmark. It is a real property of those systems on these questions, and the direction is what their designs predict, but three points do not establish that every uniform strategy must

pay this cost. The claim we defend is the weaker one: reporting a single accuracy figure would have hidden a difference that matters for choosing between them.

The separation at depth three is concentrated, and we would rather state where than have it found. 173 of the 2,649 schema-governed questions ask about the schema vocabulary rather than product content; excluding them raises the baselines to 98.99%, 99.88% and 99.39% there, and the headroom no longer rises monotonically. On that subset the baselines return a fixed premise list, consistent with the vocabulary never reaching their context, so part of the gap is an information difference rather than a reasoning one. Separately, a refusal is scored as a wrong answer, which accounts for 78% of LINC's errors. The released code reports all three, and the result that survives every re-grading is the zero in Table IV.

The verified set is, by construction, the set of questions with unambiguous answers, and its rule-governed part is checked against machine-readable constraints. A configuration that routes those questions to that check is expected to reach the ceiling. The result shows the ceiling is reachable and that the baselines do not reach it; it says nothing about the questions the set excludes. Those 645 rows are published so that this boundary can be inspected rather than taken on trust.

The one-sided result holds for these rows. It shows that on this benchmark the improvement costs nothing, but zero disagreements in 6,270 paired comparisons is not a guarantee about unseen questions.

The two arms measure different things and should not be averaged. The verified set measures accuracy where an answer exists. The public-database arm measures whether the system knows when one does not, which is why coverage and calibration are reported there. Whether 73.0% coverage is acceptable depends on what a withheld answer costs in a given setting.

Finally, this paper compares configurations. It does not attribute the difference to any single mechanism; separating the contribution of symbolic validation from that of selective abstention requires an ablation, reported separately.

VI. Conclusion

Grouping questions by how many sources an answer must combine changes what the numbers say. A single accuracy figure puts three published systems within 2.11 points of each other. Split by fusion depth, the ordering inverts: the system that formalises every question leads on rule-governed questions and trails everywhere else, while the systems that combine evidence loosely do the reverse. Reporting one number per system hides a trade-off that determines which system a practitioner should actually choose.

For product-passport records this has a direct consequence. Locating a stated value is effectively solved, so further work on retrieval has little left to recover; the available accuracy is concentrated at the depth where a retrieved value must meet the schema that classifies it. But buying that depth uniformly, by pushing every question through a formal representation, costs accuracy on the questions that never needed it. Applying validation selectively, and withholding an answer when the combined evidence is insufficient, closed the gap on our verified set without costing accuracy anywhere on it.

The domain here is product passports, whose records are schema-bound by construction, which is what makes the deepest level common. But nothing in the analysis is specific to them. Any setting where answers must combine stored records with a schema or rule base held separately from those records — clinical coding, financial reporting, safety certification — should expect the same shape, and is better served by measuring fusion depth than by reporting a single accuracy figure. Whether the pattern extends to genuinely threshold-governed compliance questions, which this benchmark does not contain, is open.

Reproducibility

The verified set, its exclusion records with reasons, the per-question outputs of all four configurations, and the public-database evaluation summary are released at https://github.com/aeturzo/dpp-qa-fusion-depth. Every number in this paper is generated from those files, and the released `verify_all.py` re-derives all of them and fails if any disagrees with the value printed here; none is transcribed by hand. The same repository reports what changes under stricter grading of the question classes.

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